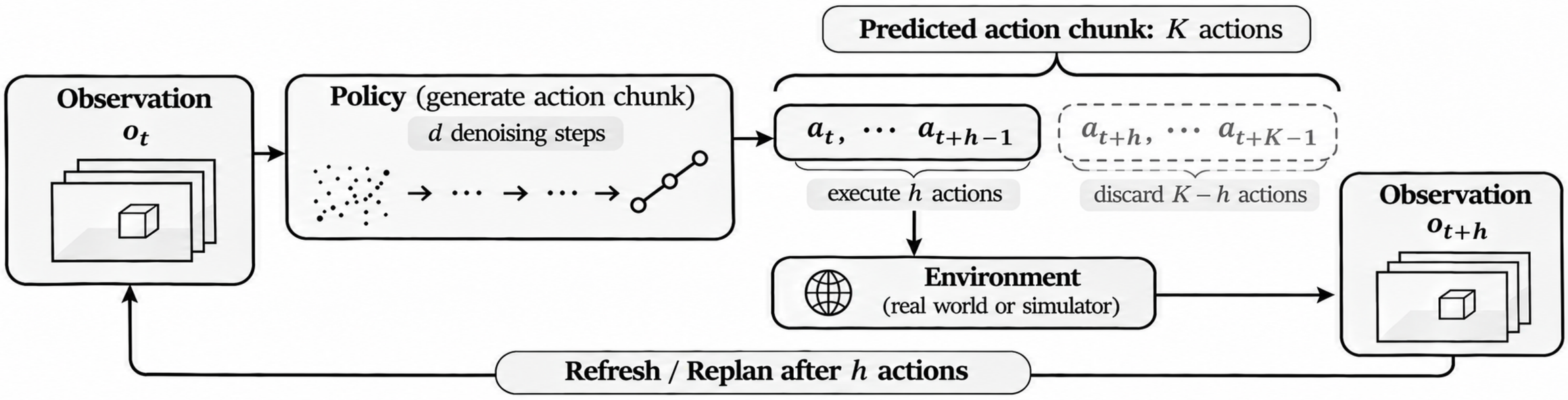
Refinement vs. Refresh

A Reproducible Evaluation Protocol for Closed-Loop Robot Policies

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Closed-Loop Controls: Refinement and Refresh



Parameters | K : number of predicted actions per chunk (chunk length) | h : executed actions per replan ($h \leq K$) | d : denoising steps (refinement)

Synchronous closed-loop execution. Each policy call predicts K actions; only the first h are executed before obtaining a new observation and replanning.

Reproducible Protocol: Separate Design, Measurement, and Evidence Status

1. Define

Fix the checkpoint and all paired rollout conditions.

2. Account

Measure the full cost of each prediction and episode.

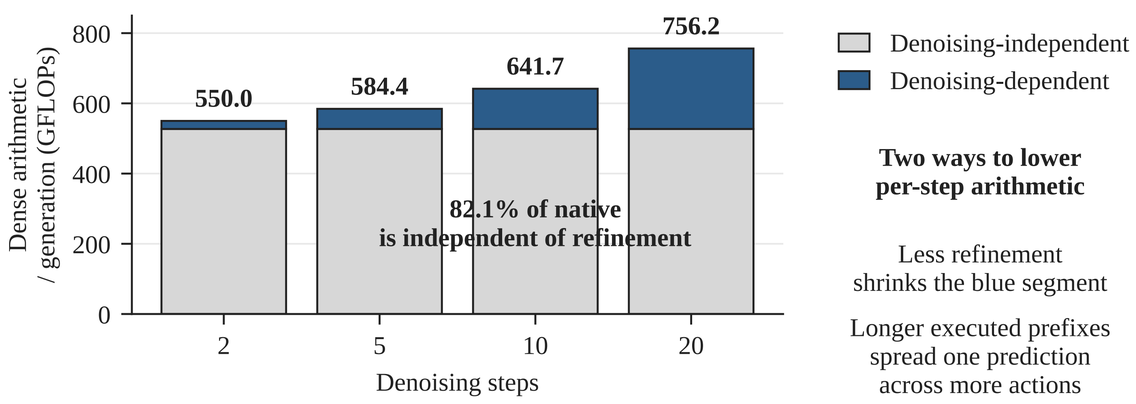
3. Freeze

Freeze primary comparisons before held-out evaluation.

4. Interpret

Label amended contrasts descriptive and limit claims.

Complete Compute Accounting

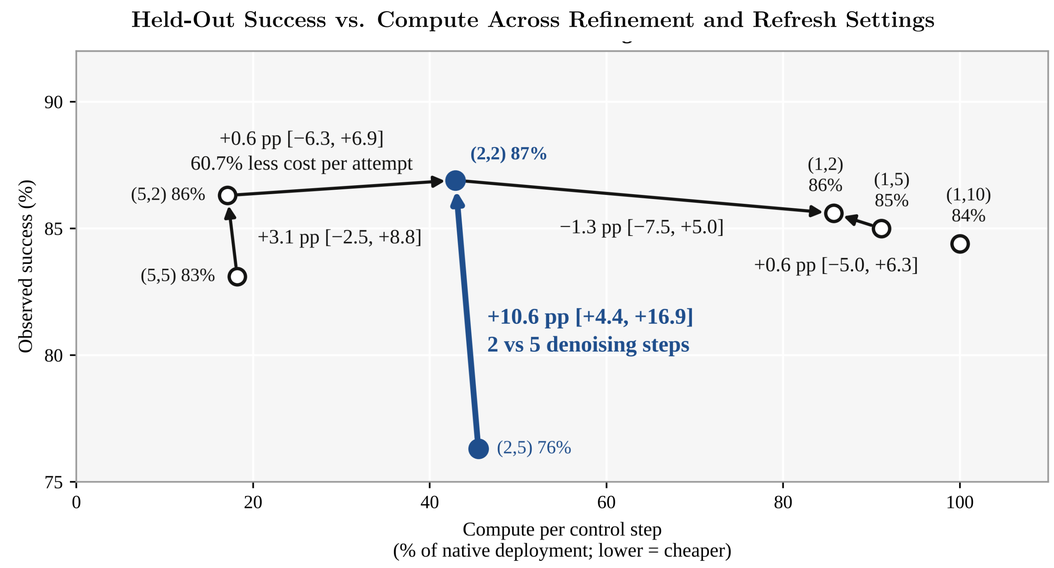


Two ways to lower per-step arithmetic

Less refinement shrinks the blue segment

Longer executed prefixes spread one prediction across more actions

Worked Example: Frozen SmolVLA



Evidence Status: What the SmolVLA Example Supports

Frozen before confirmation

At fixed refresh frequency, reducing denoising from five to two steps increased success from 122/160 to 139/160. This is the prespecified primary comparison.

Amended: descriptive only

Refreshing after five rather than two actions produced 138/160 versus 139/160 successes with 60.7% lower arithmetic per attempt. Similar observed success does not prove equal quality, and arithmetic is not deployment latency.

SmolVLA is a worked example, not a universal allocation rule. Report success and complete computational expenditure together. Keep confirmatory and descriptive evidence visibly separate.